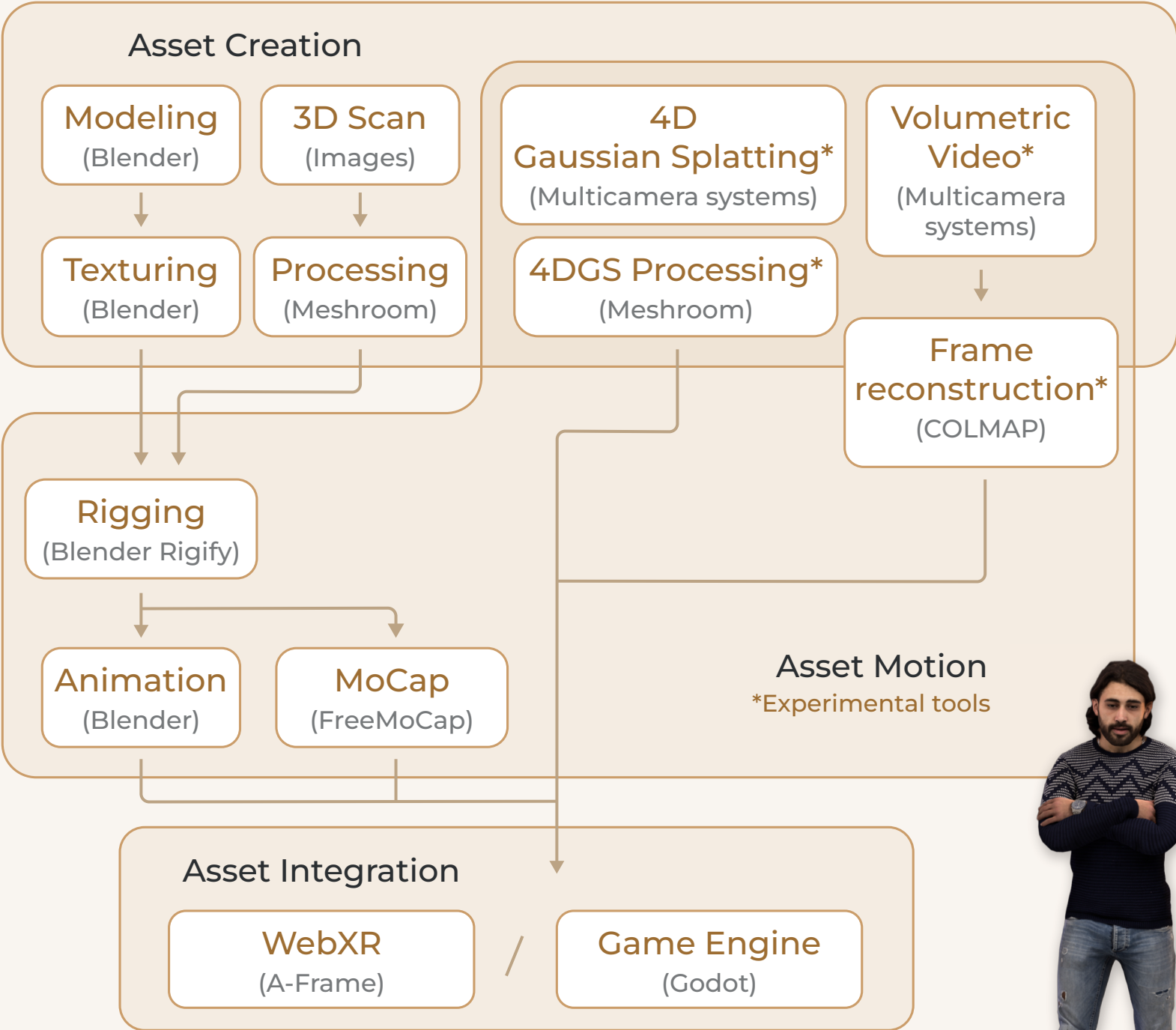


Open-Source XR Pipelines for 3D Content Creation and Integration: A conceptual workflow model

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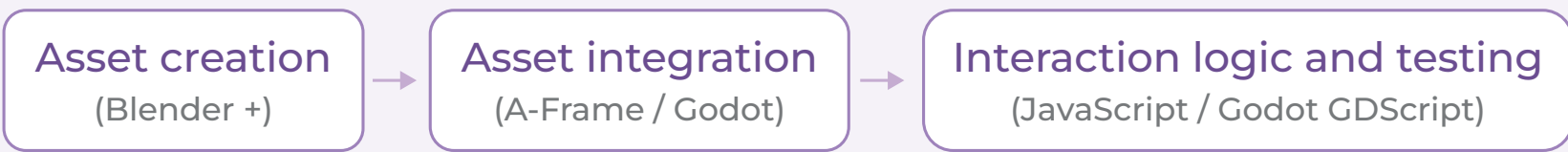
The production of extended reality (XR) content involves workflows that are becoming increasingly complex. The data sources integrated into environments within the XR medium are becoming more heterogeneous as technology advances. Among these, we can find different types that can be classified taxonomically, based on the process and mechanisms by which they were produced, their interaction with the immersive environment, or their category as ‘organic’ or ‘inorganic’. Among these process-related categories are manually created 3D assets, those resulting from the capture of an object, living being or environment present in reality, and, finally, advanced volumetric representations. These workflows are also often fragmented across proprietary tools, which limits the interoperability and reproducibility of the process. In this context, open-source ecosystems offer a flexible and scalable alternative, enabling modular workflows and transparent—albeit sometimes fragmented—processes. This paper proposes a conceptual model for an XR workflow using open-source tools, which integrates multiple types of assets through potentially interoperable formats and processing stages that are proposed for standardisation.

Organic Rigged Assets



Organic rigged assets are mainly characters integrated in a XR scene that have embedded their own animations and poses.

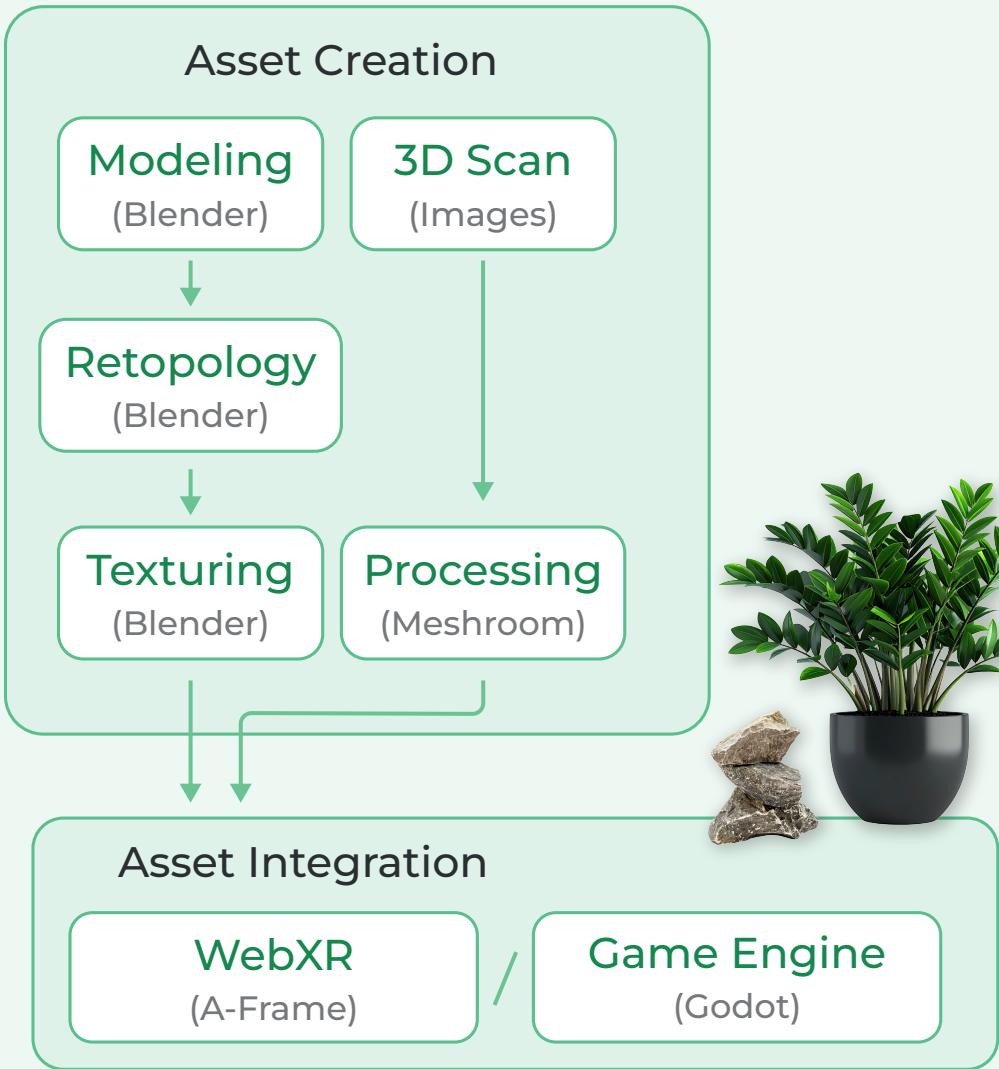
Interactive Assets (XR Objects with Triggers)



Interactive assets are those that are triggered by some specific actions, they are fundamental for enabling user engagement in XR environments.

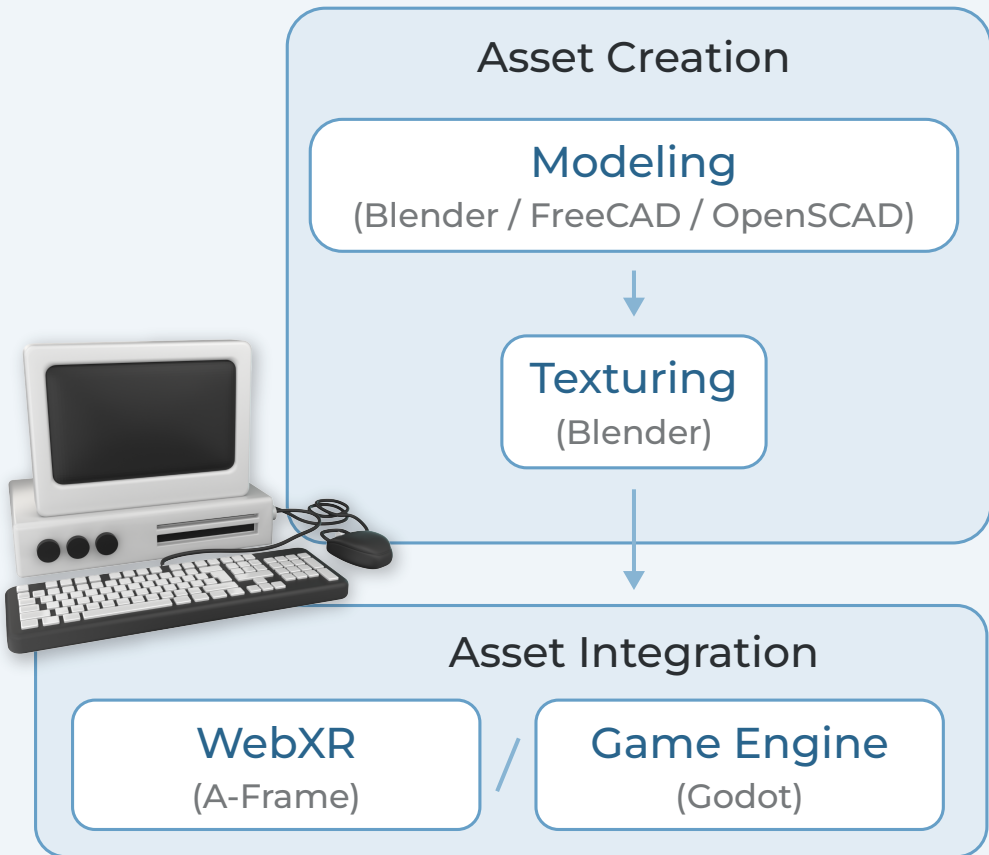


Organic Non-Rigged Assets



Organic non-rigged assets represent natural elements that do not require animation.

Inorganic / Hard Surface Assets



Hard surface assets are characterized by precise geometry commonly used for objects and architectural elements.

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